

Dengue Healthcare Analysis and Patient Outcome Analysis

Interactive Power Bi Report -2025

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533

Positive Cases

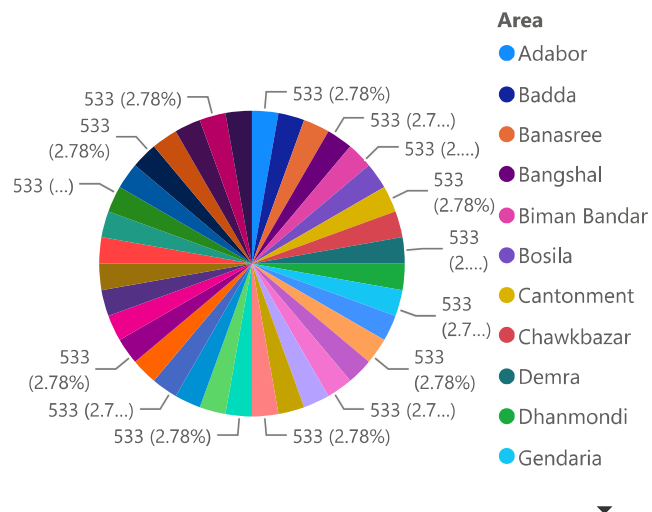
1K

Total Patients

151.72K

Average Platelet Count

Positive Cases by Area



Dengue Healthcare Analysis and Patient Outcome Analysis is a Power BI-based Business Intelligence project that analyzes **1,000 dengue patient records** containing demographic, geographic, clinical, laboratory and diagnostic information. The project studies **positive cases, symptoms, age, gender, areas, platelet count, WBC count, body temperature, fever duration and test results** to identify important patterns and trends.

Why This Dataset?

The dataset was selected because it contains multiple types of healthcare information, making it suitable for **multidimensional BI analysis, data visualization and OLAP operations**.

Benefits of the Analysis

The analysis helps identify **areas with higher positive cases, affected age groups, common symptoms and differences in laboratory results and test outcomes**. Interactive Power BI features such as **slicers, Slice, Dice, Drill-Down and Drill-Up** make the analysis easier to explore and understand.

Main Objective

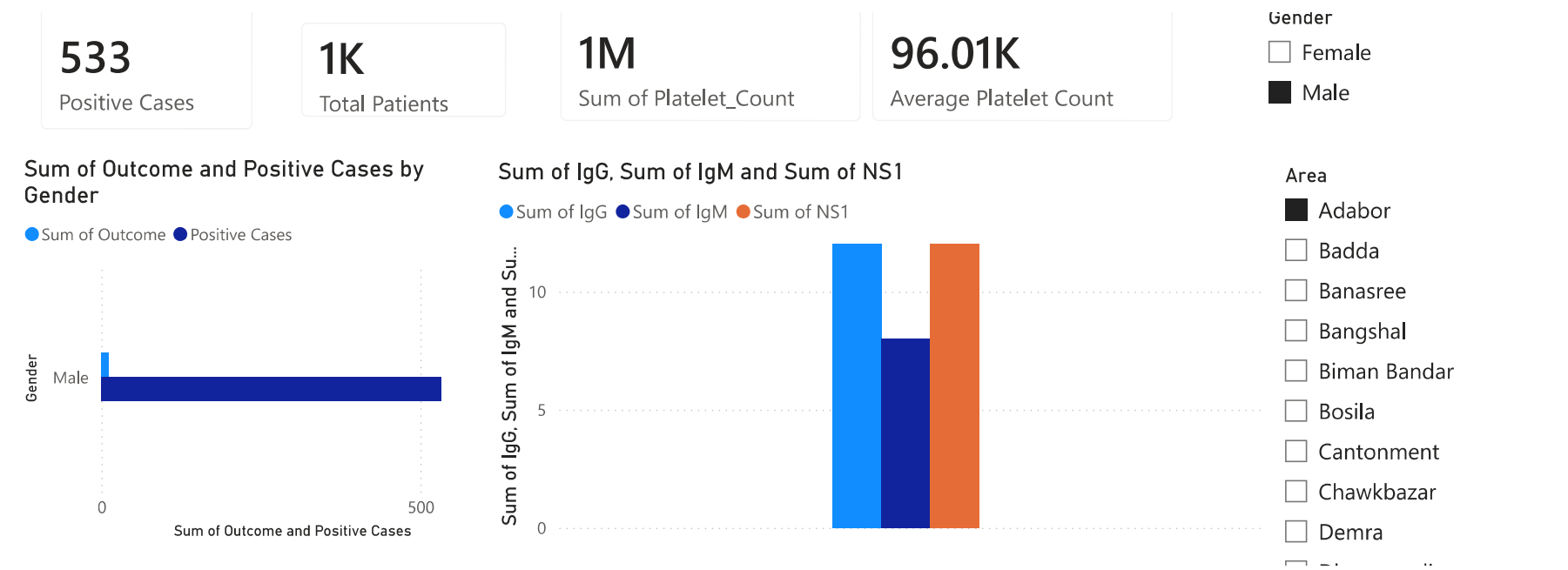
To transform dengue patient data into an interactive Power BI dashboard that provides meaningful healthcare insights and supports data-driven analysis.

Dengue Healthcare Analysis and Patient Outcome Dashboard

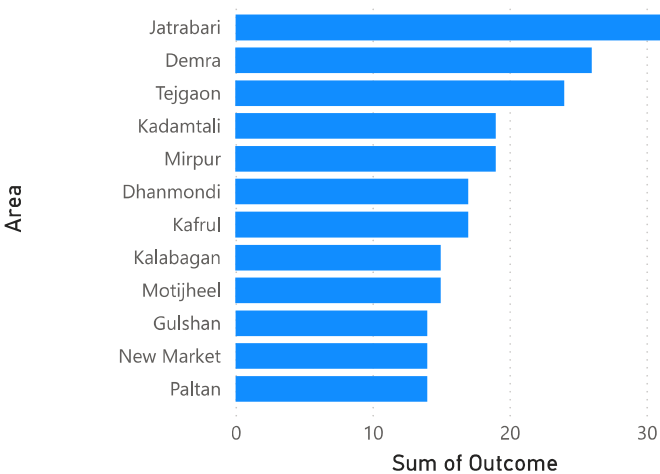
This Power BI report provides an interactive analysis of **1,000 dengue patient records**, including **demographic, geographic, diagnostic, and clinical information**. The dashboard highlights **533 positive cases** and uses KPI cards to present important measures such as **total patients, positive cases, total platelet count, and average platelet count**, giving a quick overview of the dataset.

The **Gender-wise outcome graph** helps compare patient outcomes between male and female groups, while the **diagnostic test graph** compares the recorded **IgG, IgM, and NS1 results** to provide insight into dengue-related test patterns. The **Gender and Area slicers** allow users to interactively filter the data and explore specific patient groups or locations. These features make the dashboard useful for **Slice and Dice analysis** and help users identify patterns in dengue cases more easily.

Overall, the report transforms raw healthcare data into meaningful visual insights and provides a clear view of **patient outcomes, demographic distribution, diagnostic test results, and platelet-related indicators**, supporting interactive and data-driven healthcare analysis.



Sum of Outcome by Area



- District, Area, HouseType
- ^ ☒ Dhaka
 - ∨ ☐ Adabor
 - ∨ ☐ Badda
 - ∨ ☒ Banasree
 - ^ ☐ Bangshal
 - ☐ Building
 - ☐ Other
 - ☐ Tinshed
 - ∨ ☐ Biman Bandar
 - ∨ ☐ Bosila
 - ^ ☒ Cantonment
 - ☒ Building

Dengue Geographic & Hierarchical Analysis

This page provides a detailed analysis of the **geographical distribution of dengue-positive cases** and allows users to explore the data at different levels of detail. The “**Sum of Outcome by Area**” **horizontal bar chart** ranks different areas according to the number of recorded positive cases. The chart makes it easy to identify areas with comparatively higher case counts, such as **Jatrabari, Demra, Bangshal, Tejgaon, and Kadamtali**, while areas such as **Gulshan, New Market, and Motijheel** have relatively lower values. This ranking helps in understanding the geographic concentration of dengue cases and makes comparison between areas easier.

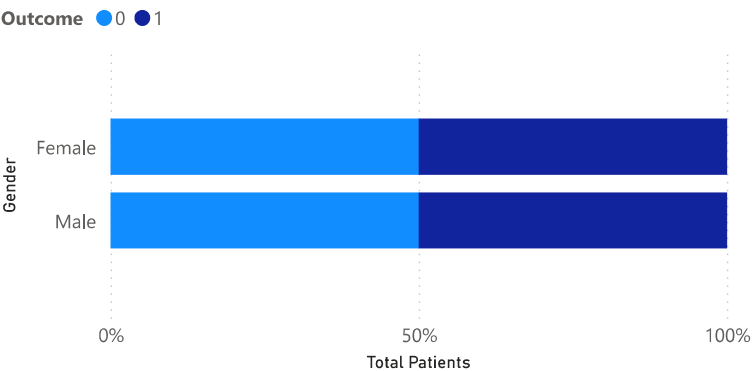
The **District–Area–HouseType hierarchy** provides a deeper level of analysis. The report begins with **Dhaka** at the district level and can be expanded to individual areas such as **Adabor, Badda, Bangshal, Cantonment** and others. Each area can then be further expanded into **HouseType categories**, such as **Building, Other and Tinshed**. This structure allows the user to perform **Drill-Down analysis**, moving from a broad geographic level to a more specific level. Conversely, **Drill-Up** can be used to move back from HouseType to Area and then to District.

The combination of the bar chart and hierarchy provides both **summary-level and detailed-level insights**. The bar chart quickly identifies areas with higher positive-case counts, while the hierarchy helps investigate the underlying distribution within a selected area. Overall, this page demonstrates how Business Intelligence can be used for **geographic comparison, hierarchical exploration and OLAP-based Drill-Down and Drill-Up analysis**, making it easier to identify location-specific patterns in the dengue dataset.

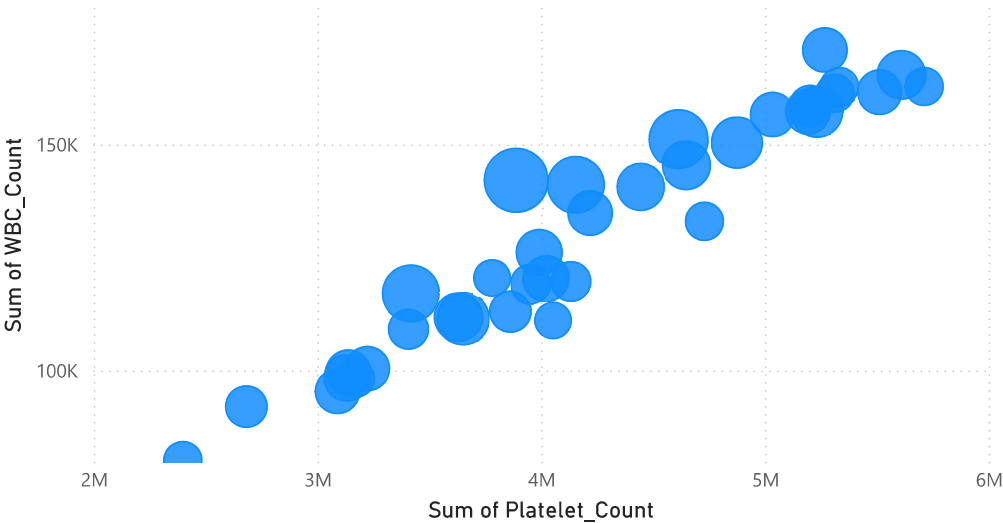
Clinical & Patient Outcome Analysis

This page analyzes **dengue patient outcomes, clinical indicators and area-wise positive cases**. The **Outcome Distribution by Gender** chart compares the percentage of positive and negative cases between male and female patients. The **clinical scatter plot** shows the relationship between **average platelet count and WBC count by area**, with bubble size representing average body temperature. The **Top 10 Areas by Positive Cases** chart ranks locations with higher dengue-positive cases, making geographic differences easier to identify. Overall, the page provides an interactive view of **gender, clinical, and geographic patterns** in dengue patient data.

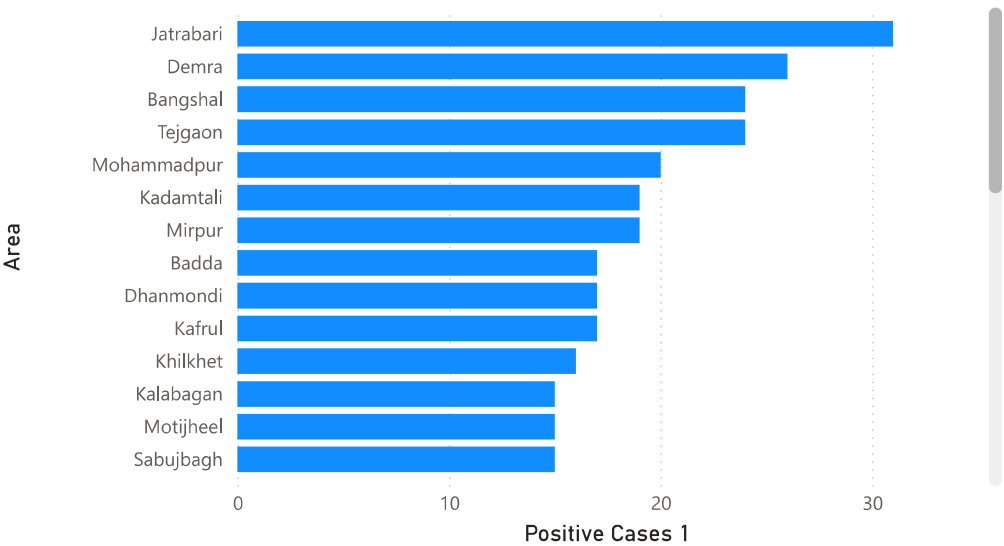
Total Patients by Gender and Outcome



Sum of Platelet_Count, Sum of WBC_Count, Sum of Outcome and Sum of Body_Temperature by Area



Positive Cases 1 by Area



Fever, Temperature & Symptom Analysis

This page focuses on the **clinical and symptom-related characteristics of dengue patients**, particularly fever duration, body temperature, headache and area-wise variation. The **Fever Duration chart** shows how fever days are distributed across patients, making it possible to observe which durations occur more frequently and compare variations in the recorded fever period. The **Body Temperature and Headache chart** provides a comparative view of temperature-related measurements and headache observations across different areas and fever-duration levels, helping to explore whether symptom patterns vary between locations.

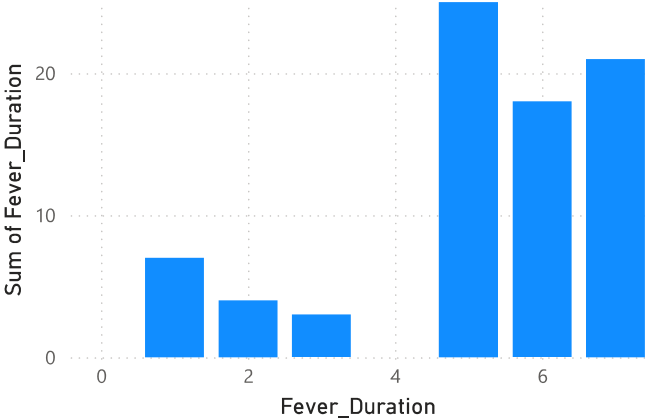
The **Area slicer** makes the dashboard interactive by allowing users to select a particular area and immediately observe the corresponding changes in the charts. This supports **Slice and Dice analysis**, as users can analyze a specific geographic segment of the patient data. The KPI at the top provides a quick summary of the selected fever-related measure and changes dynamically based on the applied filters.

Together, these visuals help users understand **how fever duration, body temperature, headache and geographic location are distributed within the dataset**. The page can also support further investigation of clinical patterns by comparing different areas and fever-duration groups, making it useful for identifying variations in patient symptoms and overall healthcare trends.

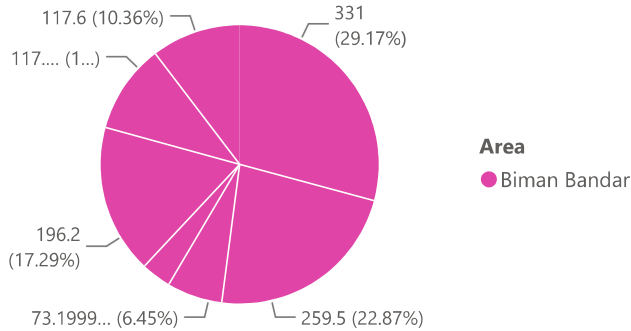
Sum of Fever_Duration by Headache

68

Sum of Fever_Duration by Fever_Duration



Sum of Body_Temperature and Sum of Headache by Area and Fever_Duration



Area

Adabor	Badda	Banasree
Bangshal	Biman Ban...	Bosila
Cantonment	Chawkbazar	Demra